

EPA Compliance Assistance Workshops Don't Reduce Violations

Lessons from “The Effect of Compliance Assistance on Pollution Discharges and Violations of Environmental Regulations” by Paul J. Ferraro & Jay P. Shimshack in *Journal of Policy Analysis and Management*, 2026.

Offering polluting facilities free help to follow the rules doesn't make them more likely to follow them. New research studying hundreds of facilities that were offered free compliance assistance workshops found the workshops didn't reduce violations.

BACKGROUND

Environmental enforcement in the U.S. has long centered on inspecting facilities, catching violations, and levying fines. **Compliance assistance** takes a different approach: regulators offer workshops, technical guidance, and other support to help facilities understand and meet regulatory requirements. This approach reflects the idea that many violations result from confusion or limited expertise rather than deliberate corner-cutting. Compliance assistance has become a prominent alternative to enforcement action at the Environmental Protection Agency (EPA), and was recently adopted by the Ohio EPA in a program offering free workshops to wastewater treatment facilities across the state.

In a new study, Ferraro and Shimshack (2026) find that compliance assistance workshops did not reduce regulatory violations or pollution discharges, despite earlier before-and-after violation comparisons suggesting that the program was effective. Such comparisons can be misleading, and accurately understanding program effects is critical to promoting environmental compliance in real-world policy settings. To measure those effects credibly, the researchers analyzed data from hundreds of Ohio wastewater treatment facilities that had recently been cited for violations and were invited to attend workshops rolled out over several months.

THE WORKSHOPS' EFFECT ON VIOLATIONS, MONTH BY MONTH

Source: Ferraro & Shimshack (2026), Fig. 2a.



Note: This figure shows the estimated effect of compliance assistance workshops on the number of facility violations. Each dot shows whether facilities had more (above the zero line) or fewer (below the zero line) violations in a given month relative to the workshop month (vertical dashed line). The estimates account for each facility's usual number of violations, the local weather, and any common trends shared across facilities; the shaded band shows the range of estimates within the 95% confidence interval. If the workshops had an effect on violations, the line would drop below zero after the workshop.

THE CHALLENGE

Measuring whether this program works requires a difficult-to-find comparison group: facilities that show what the trained facilities would have done without the training. The facilities invited to these workshops had recently been cited, and an unusually bad stretch of violations may be followed by a more typical one; violations at trained facilities may well have been on their way down before the training started. Comparing facilities that attended against those that did not raises a different problem: attendance was voluntary, so a gap between the two groups could reflect who chose to walk through the door rather than what they learned once inside.

Ferraro and Shimshack addressed this by comparing facilities trained earlier against facilities that had not yet been trained. Both groups had been invited on the same criteria, both had been cited, and both were tracked over the same months, so any trend running through the industry appeared in both. The staggered rollout of the workshops is what made the not-yet-trained facilities available as a benchmark. The comparison holds only if those facilities would otherwise have followed the same path as the trained ones, an assumption the pre-workshop months in the figure support but cannot prove.

WHAT CHANGED AFTER THE WORKSHOPS

Trained facilities were no less likely to violate environmental regulations than facilities that had not yet been trained. That held in every month the researchers tracked, not just on average: the training was neither slow to take hold nor quick to wear off.

A violation is a regulatory event rather than a measure of pollution, so Ferraro and Shimshack also measured discharges directly, across five pollutants. Both sets of estimates were close to zero and precisely estimated. If the workshops changed what facilities reported or what they released into the water, the change was too small to detect.

Violations did fall at the facilities that attended, but the workshops were not the reason. Comparing before and after their training shows a 40 to 50 percent drop — and violations at these plants were already falling before the program trained anyone.

Compliance assistance rests on the premise that facilities break the rules because they do not fully understand them. The workshops supplied exactly the technical guidance those facilities were supposedly lacking, and it made no measurable difference. **The pattern is therefore more consistent with facilities weighing the cost of compliance against the consequences of violating rather than misunderstanding the rules.**

FOR MORE INFORMATION

Ferraro, Paul J., and Jay P. Shimshack. 2026. “The Effect of Compliance Assistance on Pollution Discharges and Violations of Environmental Regulations.” *Journal of Policy Analysis and Management* 45 (3): e70056. doi.org/10.1002/pam.70056.

Paul J. Ferraro is at Johns Hopkins University. Jay P. Shimshack is at the University of Virginia. Contact Jay Shimshack (jay.shimshack@virginia.edu) with questions about this research.

The [Environmental Inequality Lab](#) is a nonpartisan research group that applies a rigorous, data-driven approach to understand how our environment shapes economic opportunity and well-being.